

POCKET SMART AI

1. INTRODUCTION

1.1 Project Overview

Pocket Smart AI – Your Smart Budget & Recommendation Assistant is an AI-powered web application designed to help students and young individuals manage their personal budgets and make informed spending decisions. The system allows users to provide their budget, preferences, and requirements, and uses Google Gemini AI to generate personalized recommendations. It helps users understand their spending needs, plan their budget, and make suitable financial decisions through a simple and user-friendly interface.

1.2 Purpose

The main purpose of Pocket Smart AI is to provide simple and personalized financial guidance based on the user's budget and preferences. The system aims to reduce the difficulty of manual budgeting, support better spending decisions, and provide AI-based recommendations that are relevant to the user's requirements.

2. IDEATION PHASE

2.1 Problem Statement

Students and young individuals often face difficulties in managing their personal budgets, tracking expenses, understanding spending patterns, and making suitable spending decisions. Manual budgeting requires time and effort, while generic recommendations may not consider individual financial situations. Therefore, there is a need for an AI-powered solution that can provide personalized recommendations based on the user's budget and preferences.

2.2 Empathy Map Canvas

The empathy map identifies the user's Think & Feel, Hear, See, Say & Do, Pain, and Gain. It helps the team understand the problems, emotions, behaviours, and expectations of students and young individuals regarding personal budget management. The analysis shows that users want better control over expenses, simple budgeting, personalized recommendations, and improved financial awareness.

2.3 Brainstorming

The team conducted a brainstorming session to generate possible solutions for the identified problem. Ideas such as AI Expense Tracking, Smart Budget Planner, AI Spending Analysis, Personalized AI Recommendations, Expense Category Management, Savings Goal Tracker, Spending Alerts, and Financial Dashboard were discussed. The ideas were grouped into Expense Management, Budget & Savings, and AI-Based Recommendations. Based on feasibility and user requirements, the most relevant ideas were selected for Pocket Smart AI.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey describes the user's interaction with Pocket Smart AI from the initial login to receiving an AI-based recommendation. The journey includes Registration/Login → Enter Budget & Preferences → Submit Requirements → AI Processing → View Recommendation → Review Details → Use Recommendation. This helps identify user needs and improve the overall experience.

3.2 Solution Requirement

The system requires a user-friendly web interface, secure user authentication, budget and preference input, AI-based recommendation generation, recommendation details, and recommendation history. The application requires React.js and Tailwind CSS for the frontend, Fast API and Python for the backend, SQLite for data storage, and Google Gemini API for AI-based recommendations.

3.3 Data Flow Diagram

The Data Flow Diagram represents how information moves through the Pocket Smart AI system. The user enters login details, budget, preferences, and requirements through the frontend. The frontend sends the information to the Fast API backend. The backend processes the request and communicates with the SQLite database and Google Gemini API. Gemini generates the personalized recommendation, which is returned through the backend to the frontend and displayed to the user.

3.4 Technology Stack

- Frontend: React.js, Tailwind CSS
- Backend: Fast API, Python
- Database: SQLite
- AI API: Google Gemini API
- Authentication: User Authentication
- Security: Environment Variables (.env) for API key protection
- Development: VS Code, Localhost

4. PROJECT DESIGN

4.1 Problem Solution Fit

Pocket Smart AI addresses the difficulty of managing personal budgets and obtaining personalized spending guidance. The system analyses the user's budget, preferences, and requirements and uses Gemini AI to generate relevant recommendations. This reduces manual effort and helps users make more informed budget-related decisions.

4.2 Proposed Solution

The proposed solution is an AI-powered web application that acts as a smart budget and recommendation assistant. Users can register, log in, provide their budget and preferences, and submit their requirements. The system processes the information using the backend and Gemini AI and provides personalized recommendations along with relevant details. The system can also maintain recommendation history for future reference.

4.3 Solution Architecture

The architecture consists of the User → React.js & Tailwind CSS Frontend → Fast API & Python Backend → SQLite Database + Google Gemini API. The frontend manages user interaction, the backend handles application logic and API communication, SQLite stores user and recommendation data, and Gemini AI generates personalized recommendations.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was planned and developed through different stages including problem identification, ideation, requirement analysis, system design, frontend development, backend development, AI integration, database integration, testing, and documentation. Tasks were divided among team members based on their skills and progress was monitored throughout the development process. The development followed an iterative approach so that features could be tested and improved progressively.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing was conducted to verify that the application responds correctly to user requests and provides AI recommendations within an acceptable time. The major functions tested include registration, login, budget input, preference submission, AI recommendation generation, recommendation display, history retrieval, and logout. The system was also checked for proper API communication, database operations, and handling of invalid inputs.

7. RESULTS

7.1 Output Screenshots

The developed Pocket Smart AI application successfully demonstrates the major functionalities of the proposed system. The output screenshots include Login/Register Page, Dashboard, Budget & Preference Input, AI Recommendation Page, Recommendation Details, Recommendation History, and Logout/Navigation screens. These screenshots demonstrate the working interface and successful integration of the application's major components.

8. ADVANTAGES & DISADVANTAGES

➤ **Advantages**

- Provides personalized AI-based recommendations.
- Helps users manage their budget effectively.
- Reduces manual effort in financial planning.
- Simple and user-friendly interface.
- Helps users understand their spending requirements.
- Can be extended with additional financial features.

➤ **Disadvantages**

- AI recommendations depend on the accuracy of user-provided information.
- Gemini API availability is required for AI recommendations.
- SQLite is more suitable for prototype/small-scale usage.
- The system does not replace professional financial advice.
- Internet connectivity may be required for AI-based processing.

9. CONCLUSION

Pocket Smart AI provides an AI-powered approach to personal budget management and personalized recommendations. The system combines a user-friendly web interface with Fast API, SQLite, and Google Gemini AI to process user requirements and generate suitable recommendations. By considering the user's budget and preferences, the application helps reduce manual effort and supports better financial awareness and spending decisions.

10. FUTURE SCOPE

The future version of Pocket Smart AI can include automatic expense tracking, bank/payment integration, advanced spending analytics, savings goal monitoring, spending alerts, mobile application support, multilingual support, voice-based interaction, and cloud deployment. The system can also be enhanced with more advanced AI capabilities to provide improved financial insights and personalized recommendations.

11. APPENDIX

- **Source Code**

```
import React from 'react';
import { createRoot } from 'react-dom/client';
import { BrowserRouter } from 'react-router-dom';
import './index.css';
import App from './App.jsx';
createRoot(document.getElementById('root')).render(
  <BrowserRouter>
    <App />
  </BrowserRouter>
);
```

- **Github and Dataset Link**

<https://github.com/bavanat/project-smart-AI>

https://drive.google.com/drive/folders/1oWEXyTGgP_qeyna9VICuEMufx6cSIIHj?usp=sharing